



30-MAVZU · III BOB. PYTHON VA MA'LUMOTLAR BAZASI

Filtrlash va shartlar. Aggregation va grouping

WHERE'da taqqoslash, AND/OR/NOT, IN, BETWEEN, LIKE/ILIKE, IS NULL; ORDER BY, LIMIT, DISTINCT; agregat funksiyalar COUNT, SUM, AVG, MIN, MAX; GROUP BY va HAVING; statistik hisobotlar tayyorlash.

Bugun nimani o'rganamiz?

- Murakkab shartlar: `IN`, `BETWEEN`, `LIKE`
- `DISTINCT` va sahifalash
- Agregat funksiyalar: `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`
- `GROUP BY` bilan guruhlash, `HAVING` bilan guruhlarni filtrlash
- Haqiqiy **hisobotlar**: sinf o'rtachasi, eng ko'p sotilgan mahsulot

WHERE operatorlari

Operator	Misol
<code>= , <></code>	<code>fan <> 'Tarix'</code>
<code>> , < , ≥ , ≤</code>	<code>baho ≥ 4</code>
<code>AND , OR , NOT</code>	<code>baho = 5 AND fan = 'Fizika'</code>
<code>IN (...)</code>	<code>sinf_id IN (1, 3)</code>
<code>BETWEEN a AND b</code>	<code>narx BETWEEN 10000 AND 50000</code> (chegaralar kiradi)
<code>LIKE / ILIKE</code>	<code>ism LIKE 'A%'</code>
<code>IS NULL / IS NOT NULL</code>	<code>email IS NOT NULL</code>

IN, BETWEEN va qavslar

SQL

```
SELECT * FROM baholar WHERE fan IN ('Matematika', 'Informatika', 'Fizika');  
SELECT * FROM baholar WHERE sana BETWEEN '2026-09-01' AND '2026-09-30';  
SELECT * FROM oquvchilar WHERE sinf_id NOT IN (1, 2);  
  
SELECT * FROM baholar  
WHERE (fan = 'Matematika' OR fan = 'Fizika') AND baho = 5;
```



AND **OR** dan **oldin** bajariladi. Qavssiz: "Matematikadagi hammasi YOKI Fizikadagi 5 lar". Shubha bo'lsa — qavs qo'ying!

LIKE — naqsh bo'yicha qidirish

- `%` — istalgan sondagi belgilar; `_` — aynan **bitta** belgi

```
SQL
SELECT ism FROM oquvchilar WHERE ism LIKE 'D%';           -- D bilan boshlanadi
SELECT ism FROM oquvchilar WHERE familiya LIKE '%ova';    -- "ova" bilan tugaydi
SELECT ism FROM oquvchilar WHERE ism LIKE '%a%';          -- ichida "a"
SELECT ism FROM oquvchilar WHERE ism LIKE '_li';          -- Ali
SELECT ism FROM oquvchilar WHERE ism ILIKE 'zarina';       -- katta-kichik harfga qaramaydi
SELECT DISTINCT fan FROM baholar ORDER BY fan;            -- takrorsiz
```

- Saytlardagi qidiruv ko'pincha `ILIKE '%so' 'z%'` bilan ishlaydi

Agregat funksiyalar

Funksiya	Vazifasi
<code>COUNT(*)</code>	Qatorlar soni
<code>COUNT(ustun)</code>	NULL bo'lmaganlar soni
<code>COUNT(DISTINCT ustun)</code>	Noyob qiymatlar soni
<code>SUM</code> / <code>AVG</code>	Yig'indi / o'rtacha
<code>MIN</code> / <code>MAX</code>	Eng kichik / eng katta

SQL

```
SELECT ROUND(AVG(baho), 2) AS ortacha, MAX(baho) AS eng_yuqori, MIN(baho) AS eng_past  
FROM baholar WHERE fan = 'Matematika';
```

GROUP BY — guruhlab hisoblash



Rangli qalamlar: aralash qalamlarni rangiga qarab uyumlarga ajratasiz (**GROUP BY rang**), keyin har uyumni sanaysiz (**COUNT**): qizil — 5, ko'k — 3.

SQL

```
SELECT fan, COUNT(*) AS baholar_soni, ROUND(AVG(baho), 2) AS ortacha
FROM baholar
GROUP BY fan
ORDER BY ortacha DESC;
```

Natija

fan	baholar_soni	ortacha
Ona tili	2	4.50
Matematika	5	4.40
Informatika	5	4.00

GROUP BY + JOIN

SQL

```
-- Har bir sinfda nechta o'quvchi
SELECT s.nomi AS sinf, COUNT(o.id) AS oquvchilar_soni
FROM sinflar s
LEFT JOIN oquvchilar o ON o.sinf_id = s.id
GROUP BY s.nomi;

-- O'quvchilar reytingi
SELECT o.ism || ' ' || o.familiya AS oquvchi, ROUND(AVG(b.baho), 2) AS ortacha
FROM oquvchilar o
JOIN baholar b ON b.oquvchi_id = o.id
GROUP BY o.id, o.ism, o.familiya
ORDER BY ortacha DESC;
```



Oltin qoida: SELECT'dagi har bir ustun yo agregat ichida, yo GROUP BY'da bo'lishi kerak

HAVING — guruhlarni filtrlash

- **WHERE** — qatorlarni **guruhlashdan oldin**; **HAVING** — guruhlarni **hisobdan keyin**

SQL

```
SELECT oquvchi_id, COUNT(*) AS soni
FROM baholar
WHERE sana ≥ '2026-09-01'      -- 1) qatorlar filtri
GROUP BY oquvchi_id           -- 2) guruhlash
HAVING COUNT(*) ≥ 2            -- 3) guruhlar filtri
ORDER BY soni DESC;           -- 4) saralash
```

SELECT'ning bajarilish tartibi

Yozilishi	Bajarilishi	Vazifasi
1. <code>SELECT</code>	5	Qaysi ustunlar
2. <code>FROM / JOIN</code>	1	Qaysi jadvallar
3. <code>WHERE</code>	2	Qatorlar filtri
4. <code>GROUP BY</code>	3	Guruhlash
5. <code>HAVING</code>	4	Guruhlar filtri
6. <code>ORDER BY</code>	6	Saralash
7. <code>LIMIT</code>	7	Cheklash

- Shuning uchun `WHERE AVG(baho) > 4` — **xato**: o'rtacha hali hisoblanmagan!

Amaliyot: hisobotlar

- **Maktab:**

- Familiyasi "ov" bilan tugaydiganlar; 2012-yilda tug'ilganlar soni
- Har fan: soni, o'rtacha, min, max
- Har sinf o'rtacha bahosi (3 jadval)
- Kamida bitta "2" olganlar (takrorsiz); top-3 o'quvchi

- **Do'kon:**

- Narxi 4 000 – 100 000 oralig'ida
- Har buyurtma summasi: `SUM(soni * narx)`
- Har mijoz: buyurtmalar soni va jami to'lov
- Eng ko'p sotilgan mahsulot; 50 000 dan oshgan buyurtmalar (`HAVING`)

Amaliyot: namunaviy javoblar

SQL

```
-- Har bir sinf bo'yicha o'rtacha baho
SELECT s.nomi, ROUND(AVG(b.baho), 2) AS ortacha
FROM sinflar s
JOIN oquvchilar o ON o.sinf_id = s.id
JOIN baholar b ON b.oquvchi_id = o.id
GROUP BY s.nomi
ORDER BY ortacha DESC;

-- Mijozlar statistikasi
SELECT m.ism, COUNT(DISTINCT b.id) AS buyurtmalar,
       COALESCE(SUM(t.soni * t.narx), 0) AS jami
FROM mijozlar m
LEFT JOIN buyurtmalar b ON b.mijoz_id = m.id
LEFT JOIN buyurtma_tarkibi t ON t.buyurtma_id = b.id
GROUP BY m.id, m.ism;
```

- `COALESCE(qiymat, 0)` — NULL bo'lsa 0

Ko'p uchraydigan xatolar

- ❌ `column "o.ism" must appear in the GROUP BY clause`
- ❌ `aggregate functions are not allowed in WHERE` → `HAVING`
- ❌ `LIKE 'a%'` katta "A" ni topmaydi → `ILIKE`
- ❌ `COUNT(ustun)` NULL'larni sanamaydi, `COUNT(*)` sanaydi

Savol

"A" harfi bilan boshlanadigan ismlarni topuvchi shart?

A `ism = 'A%'`

B `ism LIKE 'A%'`

C `ism LIKE '%A'`

D `ism IN 'A'`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

"Har bir fan bo'yicha o'rtacha baho" uchun nima kerak?

A `GROUP BY fan + AVG(baho)`

B `ORDER BY fan`

C `DISTINCT fan`

D `WHERE fan = AVG`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Guruhlarni hisob-kitobdan keyin filtrlash uchun qaysi kalit so'z?

A WHERE

B FILTER

C LIMIT

D HAVING

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

WHERE narx BETWEEN 100 AND 200 — 200 natijaga kiradimi?

A Yo'q

B Ha, chegaralar kiradi

C Faqat 100 kiradi

D Xato beradi

Javobni tanlang — to'g'ri variant yashil bo'ladi



Uyga vazifa

- 1 **Kutubxona hisobotlari:** eng ko'p o'qilgan kitob, eng faol o'quvchi, muallif kitoblari soni, kechikkan ijaralar, oylar bo'yicha ijaralar (`EXTRACT(MONTH FROM sana)`)
- 2 **SQLBolt** saytida 1–12-darslar

Yodda tuting

- `IN`, `BETWEEN`, `LIKE / ILIKE`, `IS NULL`, `AND/OR/NOT` (qavslar!)
- `DISTINCT`, `ORDER BY`, `LIMIT`
- `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`; `ROUND`, `COALESCE`
- `GROUP BY` — guruhlash; `HAVING` — guruh sharti; `WHERE` — qator sharti
- `FROM` → `WHERE` → `GROUP BY` → `HAVING` → `SELECT` → `ORDER BY` → `LIMIT`



Savollar bormi?

Keyingi mavzu: **31. Yakuniy nazorat ishi — II–III boblar (JavaScript, Python, SQL)**

Batafsil matn, misollar va testlar: **30-sql-filtr-aggregatsiya.html** darsida